

provBookR : Visualizing Data Stories

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1 Introduction

The goal of this research proposal is first to introduce and specify the definition of data provenance, review the central questions and use cases of data provenance, note various data provenance tools of increasing scope, and finally propose a new tool for end-users that makes use of existing provenance software tools in the *R programming language*¹.

Concepts— Data Provenance, Software Engineering, Information Theory, Sensory Ecology

1.1 How do you know what you know?

It is a curious thing: people know what they know because they *just do*. Rather, people know where they got the *knowledge* or *information* from because they know they read it in a book, watched an online video, or someone told them. Then, they might ask: “How do they know what they know?” One might attempt to trace the entire origin and lineage of their knowledge, but often this quickly becomes cumbersome or *intractable*. Even more, there appears to be implicit or innate knowledge that supports human’s understanding of the world, their communities, friends, family, and especially themselves (e.g., how do you know you exist?). Further still, comprehending this text, assumes the reader has also acquired some strong English linguistic knowledge and non-trivially: *What do you know when you know a language?* At any rate, although these questions have a particular philosophical flavor, these are the questions data provenance attempts to address [1, 2].

1.2 What is Data Provenance?

To know the entire history of an object’s (e.g., painting, essay, bone structure) existence (including its origin) is to know that object’s *provenance*. According to the Public Broadcasting Service (PBS), *provenance* is defined as – “*The history of ownership of a particular item. It allows the buyer to secure additional insight as to the origin of the item*”. Similarly, *Nature* defines *Data provenance* as – “*Information about the source and history of a particular data items or sets, which is generally necessary to ensure their integrity, currency and reliability.*” These histories or *stories* about the data and their custodial accounts may or may not affect their integrity. As a consequence, the maintenance of provenance plays a critical role in artistic and scientific trade, where the provenance acts as a way to track and verify an object’s history or origin.

¹<https://www.r-project.org>

1.2.1 Why Data Provenance?

In the modern world, virtually all digital communication systems deal with the processing of data in the following way:

Receive Data → **Filter Data** ↔ **Analyze Data**

Further, professional *data scientists* work with *text* data, *numeric* data, *spatial* data, and more exotic types of data and might have an even more elaborate data processing workflow:

Load Data → **Filter Data** ↔ **Clean Data** ↔ **Analyze Data** ↔ **Plot Data**

Further still, if one is interested in knowing how the data they loaded was generated (as in the case of a scientific code review), then the world would look more like this:

$$\begin{aligned} & \text{scientist}_a \rightarrow \text{Generate Data}_a \rightarrow \text{Filter Data}_a \leftrightarrow \text{Clean Data}_a \leftrightarrow \text{Analyze Data}_a \leftrightarrow \text{Store Data}_a \\ & \leftarrow \text{scientist}_b \rightarrow \text{Load Data}_a \rightarrow \text{Filter Data}_a \leftrightarrow \dots \leftrightarrow \text{Store Data}_b \leftarrow \dots \\ & \leftarrow \text{scientist}_n \rightarrow \text{Load Data}_\phi \rightarrow \dots \leftrightarrow \text{Store Data}_n \end{aligned}$$

Here, *scientist_a* generates the initial data, *scientist_b* receives this data from *scientist_a*, and *scientist_n* retrieves their data from somewhere in the data lineage ϕ . Indeed, it is common for scientists to receive their data from groups of scientists who are from within or outside of their research institutions, government agencies, industry labs, or programming communities. Of course, scientists may even retrieve data directly from individuals and/or even themselves during close collaborations.

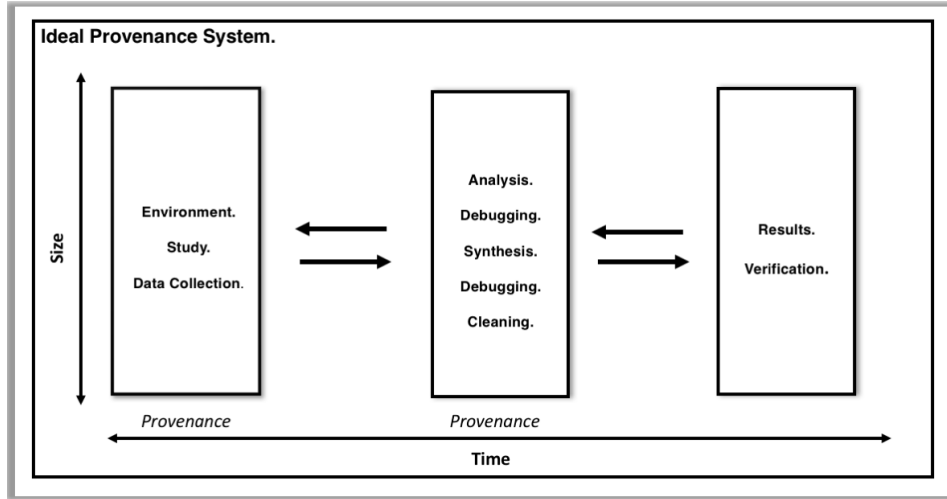


Figure 1: A *Ideal Provenance System* where the provenance metadata (Environment, Study, Data Collections) and scientific metadata (Data Analysis, Data Debugging, Data Synthesis, Data Debugging, Data Cleaning) is collected iteratively. One might also collect provenance over results and verification procedures of the experiment, but often, the results and verification processes are coupled (dependent) to the metadata. That is to say, the same metadata or initial conditions parameters should produce the same results. Finally, assuming the provenance system records the full scope of data, the size of the provenance increases with time.

Certainly reasoning about historical dependencies of particular data objects and entire scripts can be difficult. However, when the appropriate data custodial work has been completed, *metadata* allows users to answer basic questions:

- (a) *Who collected the data?*
- (b) *Where was the data collected?*
- (c) *What instruments were used?*
- (d) *What was the weather like?*

On the other hand, answering, *how has the data been manipulated?* can be done with with *scientific processing metadata* and data provenance. More specifically, data provenance helps users answer questions of the form: [2, 3]:

- (a) *What software or Web services were used to compute the data?*
- (b) *What was the input to the software?*
- (c) *Which version of my script did I use?*
- (d) *Where did input come from, etc.?*
- (e) *How are errors in sensor readings or computations identified and handled?*
- (f) *Why were the data collected?*

1.3 A Brief Natural History of Data Provenance

In the previous section the value of collecting provenance has been outlined. In particular, data provenance collection is capable of addressing a number of philosophical (“How do they know what they know?”), scientific (“What was the weather like?”), and technical (“What software or web services were used to compute the data?”) questions about data and metadata. These questions or general *use cases* have taken shape over the many years since Becker and Chambers initial description of a provenance system [4] and ultimately translated into data provenance tools being theorized and implemented in software. Example of such implemented software include, but are not limited to *operating system scope* data provenance tools like *PASS* (Provenance-Aware Storage Systems)[5], *workflow scope* data provenance tools like *Kepler* [6] and *Taverna* [7]. Smaller scale (*file scope*) data provenance tools in *R* and *Python* such as *Markdown* and *Jupyter notebooks* [8], *rctrack* [9], *ProvDB* [10], *noWorkflow* [11], *yesWorkflow* [12], and others also help with answering some of the posed questions and provide methods for data synthesis and data analysis reproducibility.

1.3.1 End-to-end Data Provenance

Still, however, it is possible to collect provenance at even more localized (system) scope, which the *R* package *RDataTracker* (rdtlite)² accomplishes without directly modifying the *R* interpreter and storing the provenance in a *Data Derivation Graph* (DDG) [3, 13, 14]. This scope of data provenance collection aids *R* data scientist and even hobbyist *R* programmers go from collecting data lineage from the workflow scope to the *language scope*. In general, the combination of data provenance collection from adaptive complex data processing systems at varying scales can allow users to collect provenance from *End-to-end*, where both ends may be characterized by dynamic granularity across *size* (space), *time*, or *complexity* [15, 16].

²*End-to-End-Provenance* (E2E) project on GitHub: <http://end-to-end-provenance.github.io/>

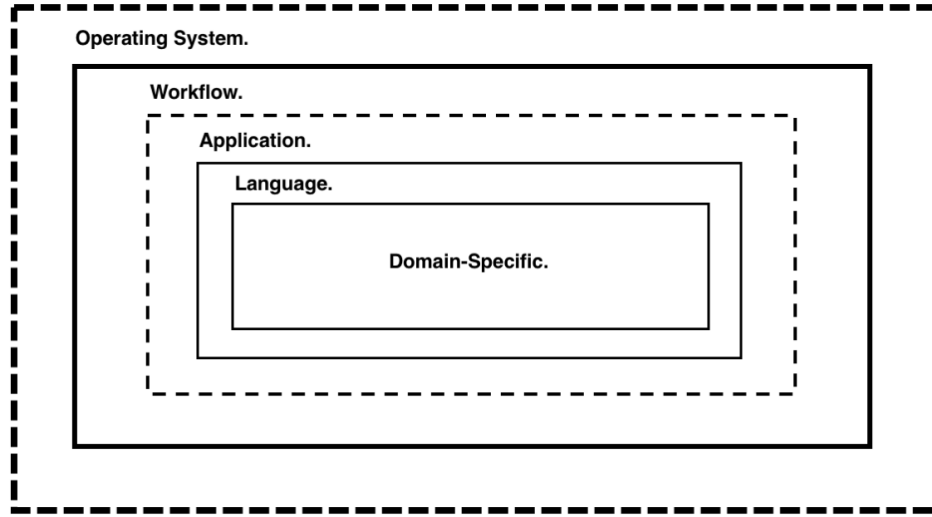


Figure 2: Scopes of provenance collection scope, where each scope level may differ in the level of provenance granularity collected.

2 Proposed Objectives & Methods

The proposed research program will start with collecting user requirements through informal mental and written notes with scientists at the professional (faculty), graduate, undergraduate, and citizen (staff) level at *Harvard University’s Long-Term Ecological Research site (LTER), Harvard-Forest*³. These notes will not be associated (identifiable) with any particular individual or subject to subsequent data analysis. These notes will only serve as guiding design principles.

Name of Use Case: `visualizer()`

Actor(s): DataScientist

Flow of events:

1. DataScientist writes an R script
2. DataScientist collects provenance using provenance tools.
3. DataScientist specifies the data/provenance they would like to visualize
4. DataScientist generates digital booklet via **provBookR**

Figure 3: Example use case: *functions* and *features* are described formally in terms of *actors*, and *event flows*.

Once “notes” have been collected, these requirements will be processed to generate use cases in the form of extended notes, flow diagrams, and other representations as is common practice in *software engineering* [17] and *product design* [18] (see **figure 3**). These notes will serve as the guiding principles for the design of the proposed software architecture’s form and function. The form and function will be specified graphically using *graph theory*, as shown in figure 4, and the various functions will

³Harvard-Forest is a highly interdisciplinary research institution focused on the understanding, management, and conservation of New England forest ecosystems

be formally described using an information-theoretic approach. In principle, because data *is* information, this allows both the abstraction of common data provenance operations supported by tools within the E2E repository (i.e., *data provenance recording*, *data provenance querying* and *searching*, and *data provenance processing* as shown in **figure 4**). These notes will serve as the guiding principles for the design of the proposed software architecture’s form and function. The form and function will be specified graphically using *graph theory*, as shown in **figure 5**, and the various functions will be formally described using an information-theoretic approach. In principle, because data *is* information, this allows both the abstraction of common data provenance operations supported by tools within the E2E repository (i.e., *data provenance recording*, *data provenance querying* and *searching*, and *data provenance processing*).

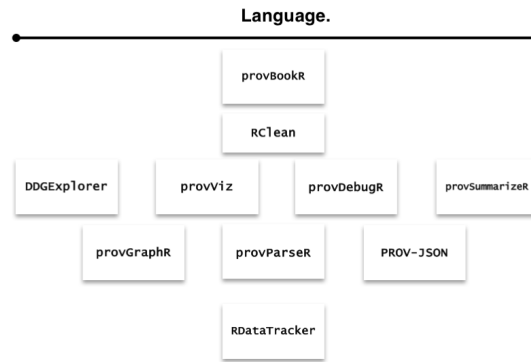


Figure 4: The current collection of E2E R provenance tools in R. *provBookR*’s software architecture will make use of these provenance packages.

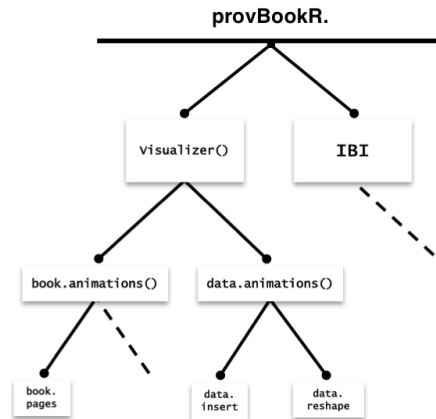


Figure 5: Graphical description of *provBookR*’s form (hierarchical) and functionality.

2.1 Proposed Software Architecture

The software architecture will make use of the existing E2E tools. Concretely, the architecture proposed being built is called *provBookR* and is a web application generated from within R. The application (APPLICATION) is composed of a “front end” (FRONTEND) and a “back end” (BACKEND).

The BACKEND architecture is developed in R and expects as input a file written in R or a JSON (JavaScript Object Notation) provenance file (PROV-JSON) (see **figure 8**). From input to output, the E2E tools acts as middleware; handling the *recording*, *querying*, and *processing* of the provenance. The output of the BACKEND is a parsed JSON file, which is input into the FRONTEND. The output of provBookR’s FRONTEND is the content the users see which is generated with *HTML*, *CSS*, and the *JavaScript* programming languages (see **figure 6:9**). As a multimedia web application, provBookR makes use of a variety of data types (font, image, text, video). Accordingly, the page animations of provBookR is handled with Javascript plugins *JQuery* and *BookBlock*. Because the provBookR works on top of the E2E end-user tools (i.e., *provSummarizeR*, *provDebugR*, *provViz* and *DDGExplorer* as shown in figure 4), the last proposed feature is to animate the data lineage to users, which could support the understanding of how data objects change through data *insertions*, *rearrangements*, *synthesis* and other data *transformations* and *augmentations*.

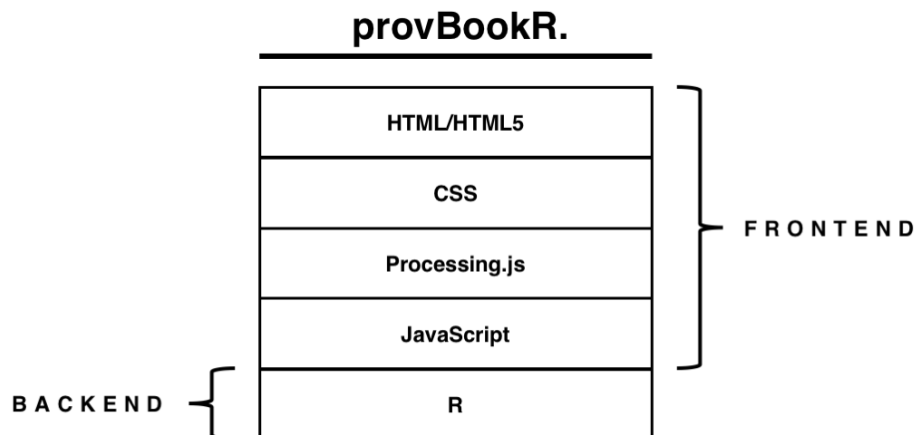


Figure 6: Abstracted FRONTEND and BACKEND layers of provBookR.

2.2 Extended and Advanced Features

In the previous section, the underlying architecture and associated features of provBookR were specified. Here, we propose extended features that would enhance the package’s utility.

2.2.1 Layout Specification

Users being able to specify the layout of their provbook is a particularly useful feature. For example, a user would have the ability to specify the *page layout*, type of *page animations*, and/or *duration* of components. This feature supports the individual’s ability to *design* provBookR’s output document, efficiently extracting relevant information, both in terms of provenance and knowledge transfer.

2.2.2 Information-Theoretic Metrics

The provBookR package views information and data as the same entity. In doing so, we vision the advanced feature that allows users to capture information-theoretic metrics directly from the prove-

nance. For example, the user might want to specify an information gain or loss comparison between two provenance files (provenance graphs), which are generated from recording the provenance on an R script at two distinct times. Hence, information metrics like *Shannon's Entropy* – $H(X)$, *Conditional Entropy* – $H(X;Y)$, *Mutual Information* – $I(X;Y)$, *Relative Information* – $D(p || q)$ [16], and derived information theory metrics like the recently proposed *Information-Balance Awareness* (IBI) [15] could be collected and further analyzed for helping answer questions related to the script's lineage.

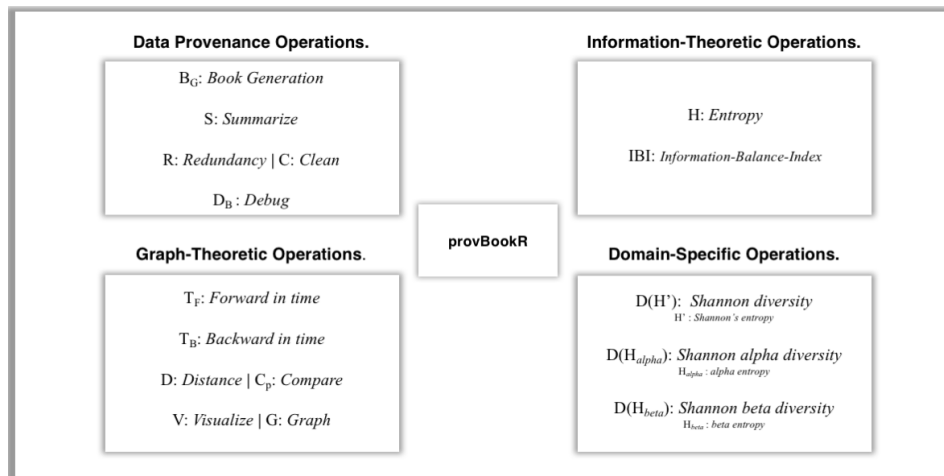


Figure 7: Extended analysis architectures provBookR could support in the future.

3 Conclusion

Data provenance has promised and provided various tools for data synthesis and reproducibility within many scientific fields. In particular, the use of data provenance in ecological studies has enabled the generation of novel methods for improving the confidence in data acquired in the field. Researchers in ecology, a notably noisy (i.e., natural environment, statistical analysis) and challenging field of scientific inquiry have conjectured about the utility of End-to-end data provenance frameworks. Accordingly, this research project proposal outlines the construction of an even friendlier framework that makes use of fine-grained data provenance tools in R. provBookR is a data provenance package that aims to support the E2E promise and encourage to use of collecting data provenance by allowing users to visualize the lineage of their data, as well as, associated data transformation animations. Using the above-mentioned R data provenance tools, JavaScript, CSS, and HTML, scientist and software engineers can generate clear visualizations of a data objects lineage.

4 Discussion : Novel Narrations in Data & Nature

As part of this research project proposal's overstory are the parallels (*isomorphisms*) between disciplines. For example, there are several relations between biology and sociology (or culture and societal studies). Just as epidemics, videos, songs, and tweets can go "viral" and in this progressively interconnected age, the culture (e.g., homogeneous information, a collection of news and media

stories) and how that culture gets transmitted (e.g., communication systems, oral traditions, media outlets, and social media) provide a unique window into understanding the past, current, and future health of a community or verification, “debugging” and reproduction of statistical analysis on biological and ecological investigations. On the other hand, mathematical models and software tools do not save lives alone. At best they reliably capture data, which is modestly a snapshot of the world, or a human expert’s intuition about how the world *should* work. In a way, these mathematical formulations and the data that drive them are also, stories. Nevertheless, without the engineering, recording, analysis, and presentation of these mathematical and computational, medical and biological, journalistic and socio-cultural stories, *proofs, errors, cures, and improvements may go undocumented and the fruits of provenance unseen.*

```
# Users would load their provenance packages
>library(rdtlite)
>library(provSummarizer)
>library(provViz)
>library(provBookR)
.
.
.
# Users might want to visualize their provenance
>prov.visualize.file(prov.file)
.
.
.
# Users might want to summarize their provenance
>prov.summarize.run("myscript.R")
.
.
.
# Users might want to make a book out of their provenance
>provBookR(prov.file, c(H, D, S, B))
>provBookR("myscript.R", c(H, D, S, B))
```

```
# Users would load their packages
library(ggplot2)
library(provBookR)
library(tidyverse)
.
.
.
.
.
.
# Generate a figure using ggplot
ggplot(data = my.data.frame, mapping = aes(x = dist, y = delay)) +
  geom_point(aes(size = count), alpha = 1/3) +
  geom_smooth(se = FALSE)

# Call provBookR
provBookR(my.data.frame, c("H", "D", "S", "BG"))

# The provBookR parameter or operation list are case-insensitive
provBookR(my.data.frame, c("H" | "h", "h.beta" | "H.BETA" | "h.BETA"))
```

Figure 8: The first code snippet is an example of users interfacing with the E2E provenance tools in R. The second code snippet is an example of how users might interface with the provBookR library (second snippet). In the second code snippet, the users can perform data operations, which provBookR uses to generate the provbook.

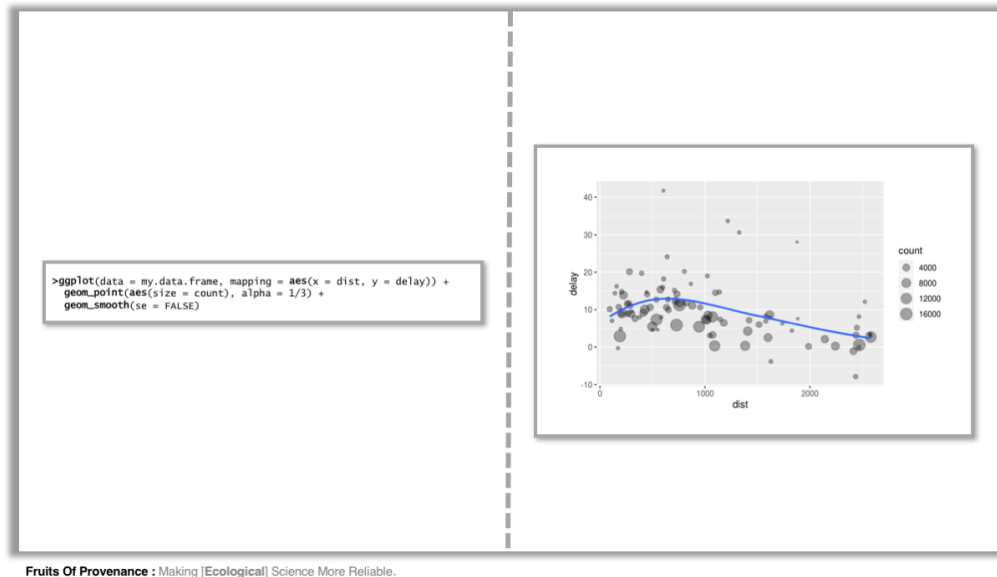


Figure 9: Example of a provbook (or *prov book*) the provBookR library would generate for the user. Here, two of the pages represent a particular line of execution in the code. The presentation of the code is on the left page, and the data visualization and animation are on the right.

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